Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy in GI Oncology

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Today’s argument

> **New RAS therapies need new kinds of trials.**

Methods to optimize combination doses are established. Designs that adapt to emerging resistance are enrolling.

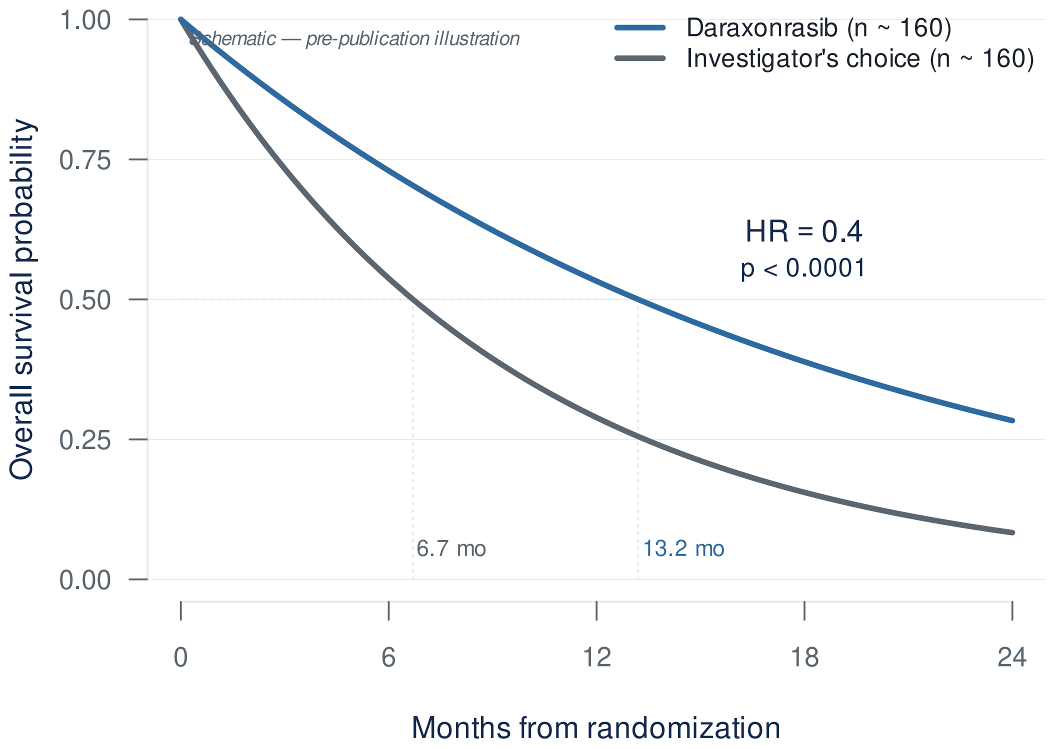
> **The difficulty: getting them calibrated and launched.**

The clinical case

Why combinations, and why now?

The RAS revolution has arrived for PDAC

- **RASolute 302** (2L mPDAC, *NEJM* 2026)[1]:
  - **Median OS: 13.2 vs 6.7 months**
  - **HR 0.40 — 60% reduction in risk of death** (p < 0.0001)
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



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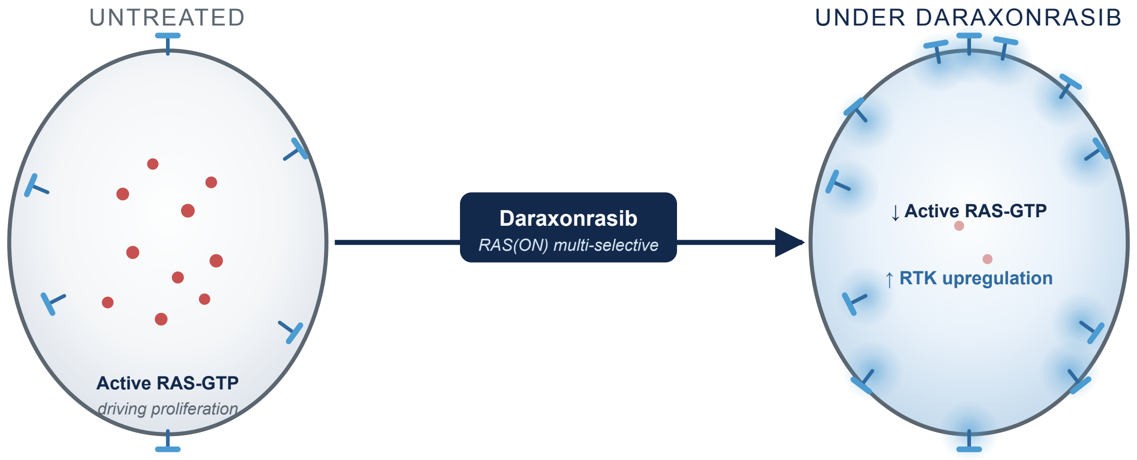
…but monotherapy has visible ceilings

RMC-6236-001 (PDAC cohort, NEJM 2026)[2]

> **Half of patients on monotherapy at 300 mg already need dose adjustments.** Combinations layer on top — with less room to spare.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations[3]
- **1L combination cohort** (daraxonrasib **200 mg** + GnP, n=40)[4]:
  - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



RTK upregulation is simultaneously a resistance mechanism and a rational combination target

The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated[5,6].
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics[7], it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension — combinations strain all three.
- **No joint-surface model.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the two-drug surface is never characterized.

The Amgen case: sotorasib + pembrolizumab

Same sponsor, two different designs:

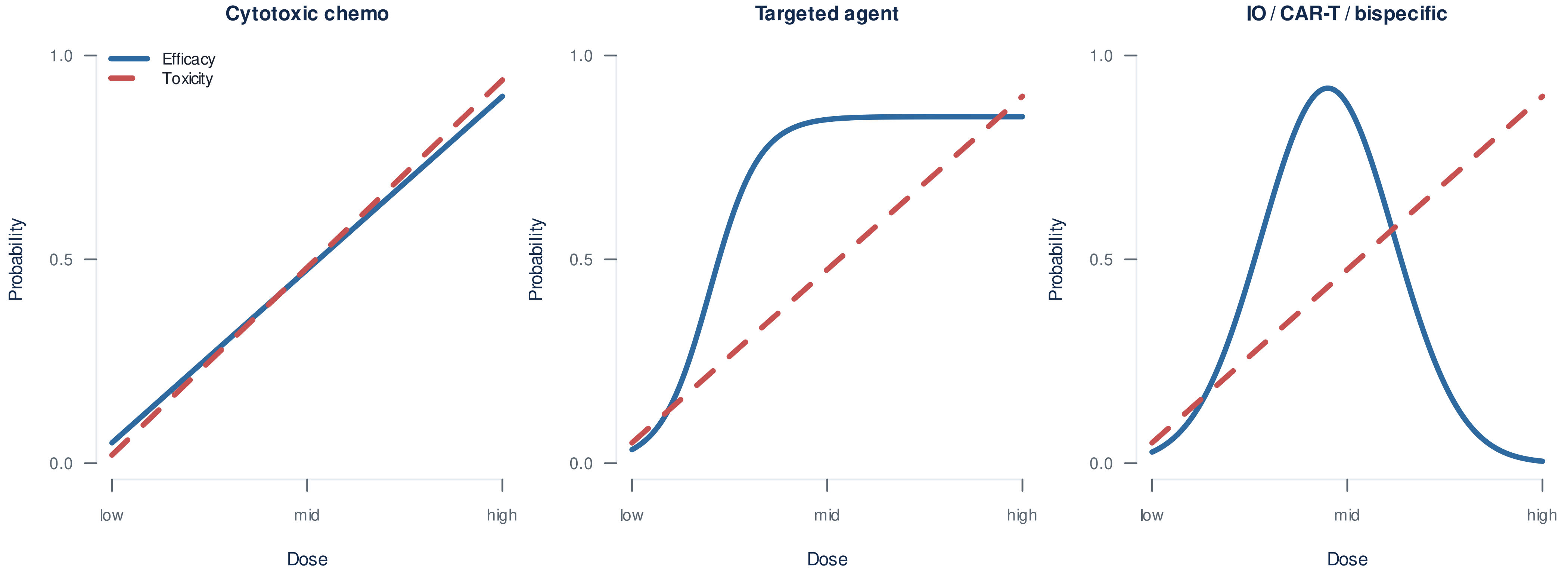
- **Monotherapy (CodeBreaK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window[8]

> **What MTD-only dose-finding was never built to handle, all at once:** joint surface unsearched, non-additive toxicity, late-onset toxicity.

Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies



The cost of MTD-only dose-finding

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)[10]:
  - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
  - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure[11]
- **MTD ≠ optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions[12]

Same drug class as daraxonrasib.

Project Optimus: the regulatory ground has shifted

How OBD selection actually works

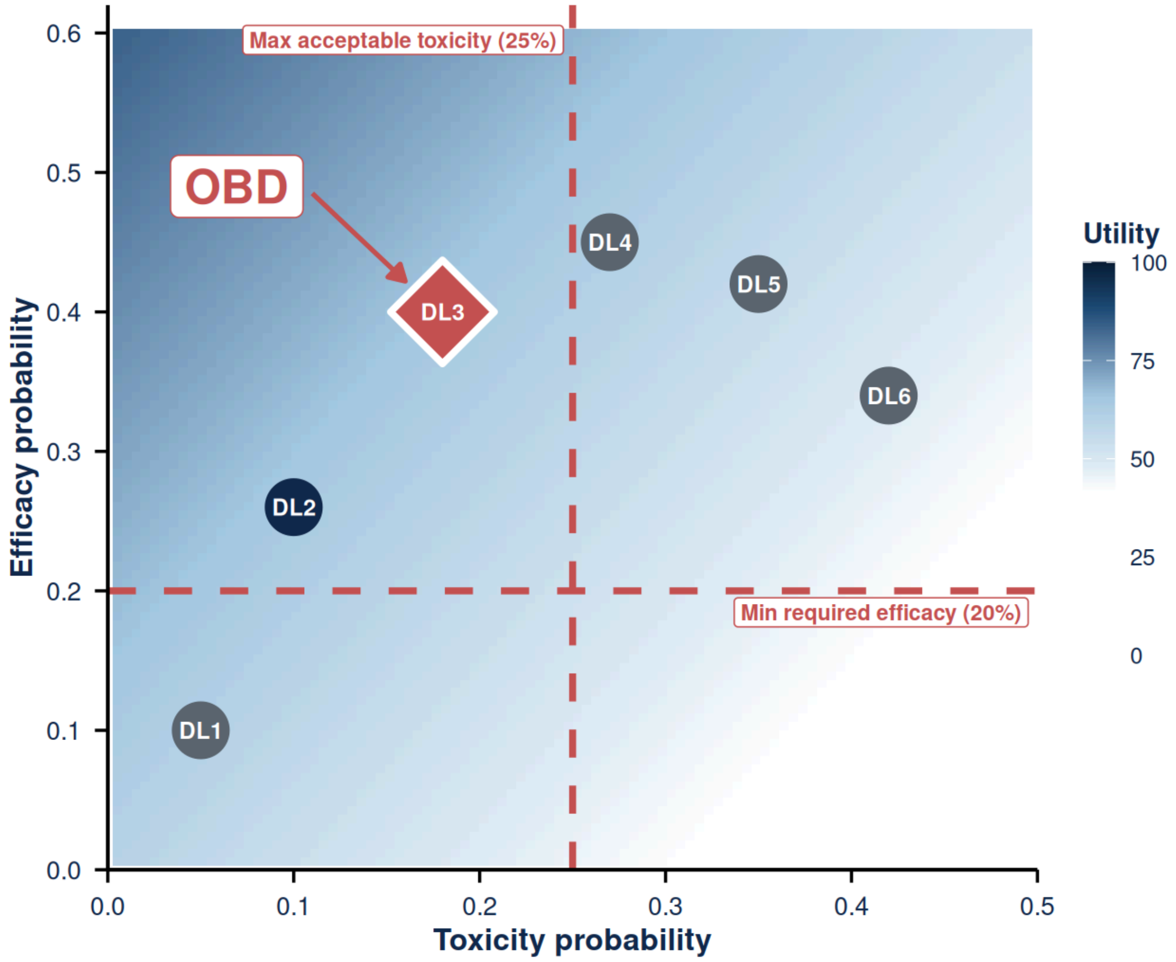
Utility-based dose-finding, in one sentence.

"A response is worth this; a Grade 3 toxicity costs this."

The team writes that utility table down up front. Bayesian, model-assisted design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)[14]:

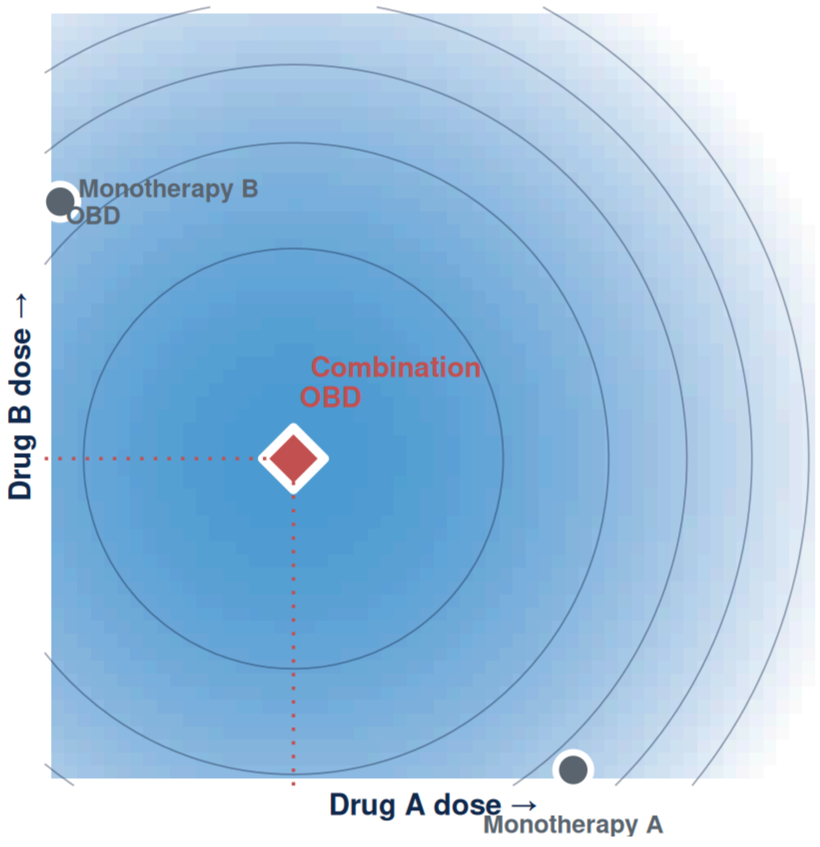
	TOX	NO TOX
Response	40	**100**
No response	**0**	60



OBD = admissible dose (inside red lines) with the highest utility.

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

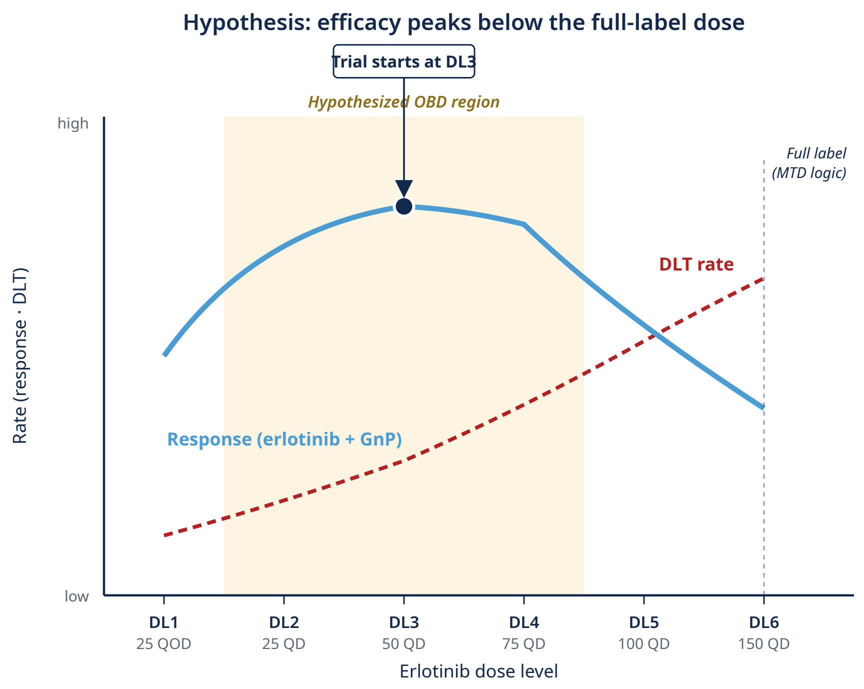
- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

> **Fix what you know. Search what you don't.**

RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface[4].

Finding the right erlotinib dose in basal-like PDAC

- **Trial:** LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by **PurIST**[17]
- **Combination:** erlotinib + biweekly gemcitabine/nab-paclitaxel
- **Hypothesis:** lower-dose erlotinib may be more efficacious (OBD-below-MTD)[18]
- **Six dose levels:** 25 mg QOD → 25 / 50 / 75 / 100 / 150 mg QD; **start at DL3**

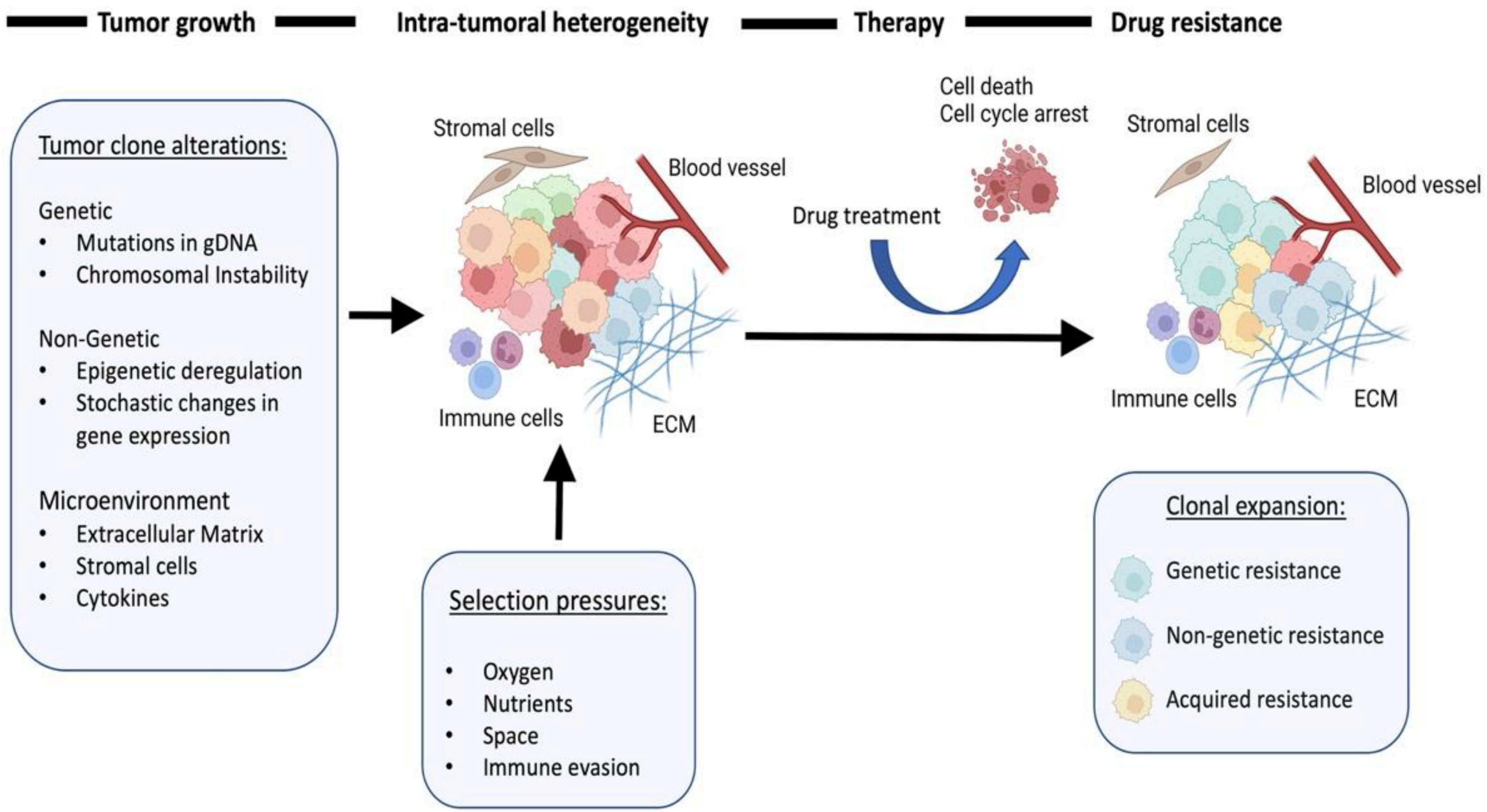


Design: *Stage 1: establish safety envelope*[19] (target DLT ≤25%, cohorts of 4). *Stage 2: find the dose with the best response/toxicity trade-off*[14] (tox ≤25%, eff ≥20%). **Up to 52 patients; ~28 at OBD** (Scenario 1).

Resistance is the next problem

Phase 2 onward: *each patient is a moving target*

Resistance is a dynamic process

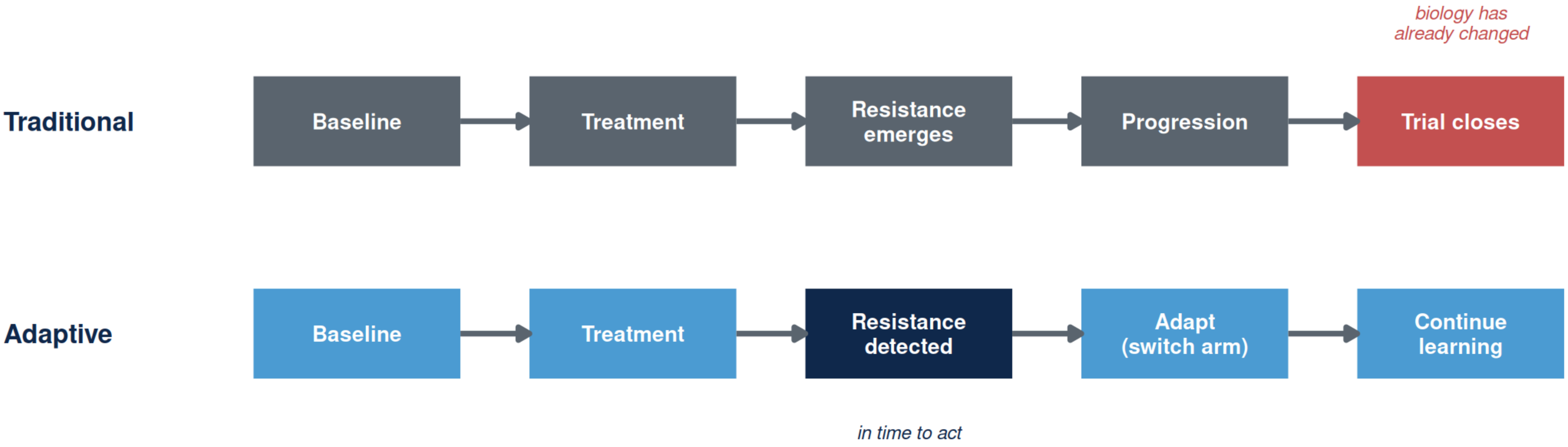


Adapted from Karagiannis & Rampias, Cancers 2022[20].

Resistance mechanisms emerge under RAS-inhibitor pressure

> **Daraxonrasib monotherapy in mPDAC** (n=44, paired ctDNA): **59% show RAS-pathway resistance at progression — KRAS amplification in 36%**[21]

Why classical trial designs miss resistance



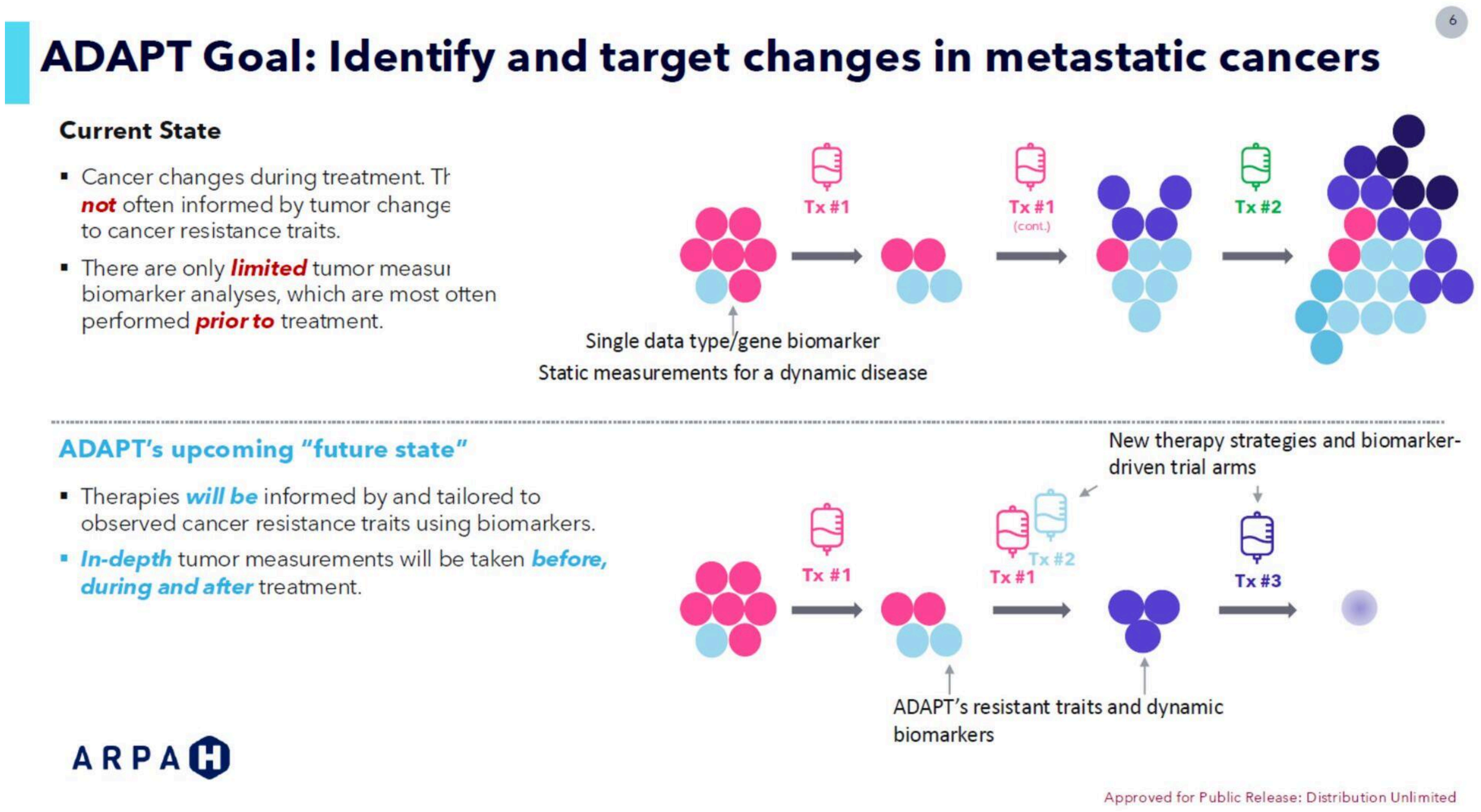
> **Genomic → ctDNA · non-genomic / plasticity → biopsies, transcriptomics, imaging.** A resistance-aware trial has to **do both** during enrollment[22,23].

ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

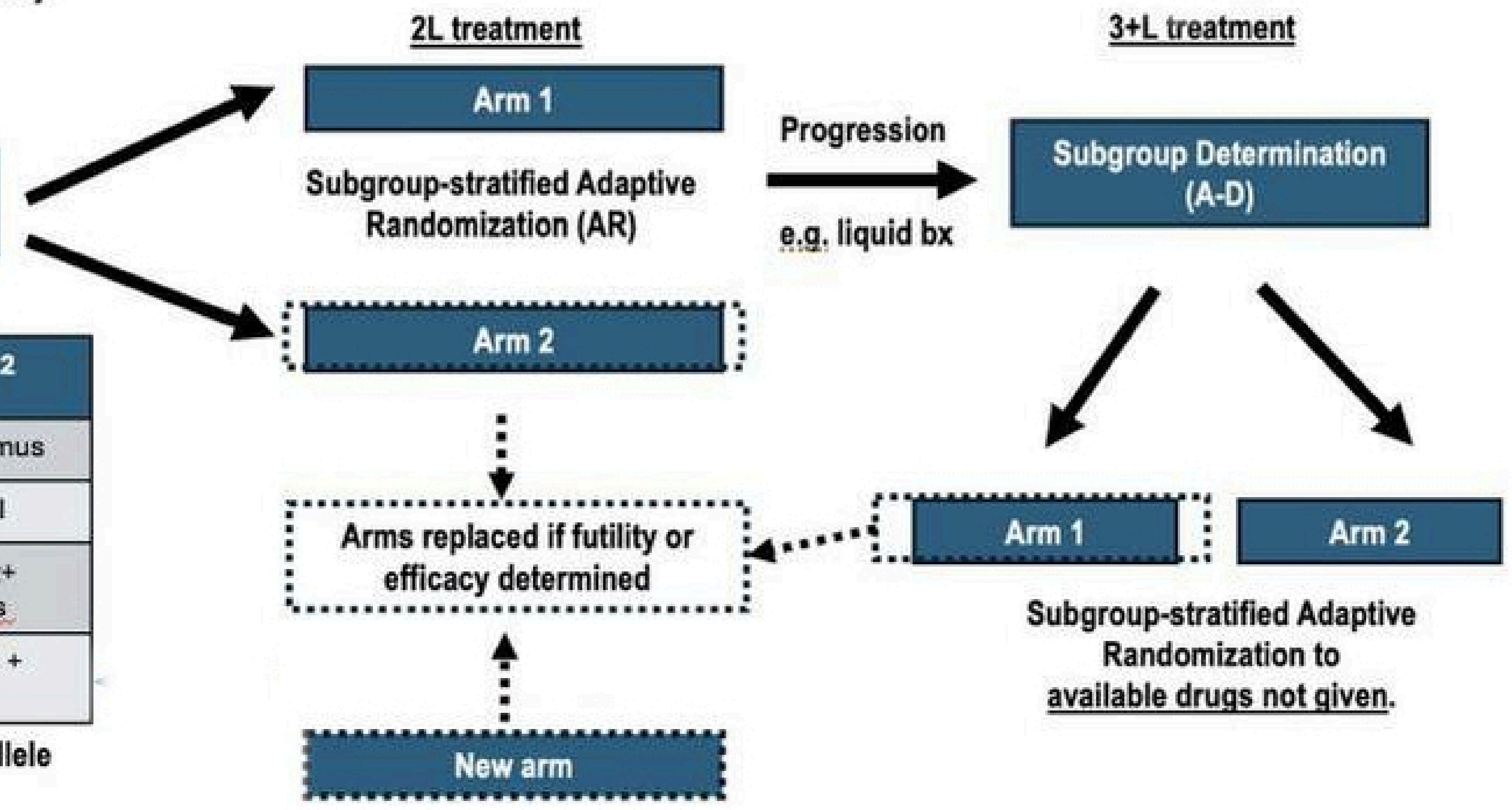
- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

ARPA-H ***ADAPT*** *→* ***EVOLVE*** *(UNC-led; Carey MPI, Rashid MPI).* ***Enrolling.***



ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

EVOLVE-BDT — and what this would look like in PDAC



ER+HER2- Initial molecular subgroups* and treatment candidates

Subgroup	ESR1 Mut	PI3K/AKT/PTEN Mt	Initial Arm 1	Initial Arm 2
A	-	-	FUL + CDK4/6i	FUL + Everolimus
B	-	+	FUL + CDK4/6i	FUL + CAPI
C	+	-	Elacestrant + CDK4/6i	Elacestrant+ Everolimus
D	+	+	Elacestrant + CDK4/6i	Elacestrant + CAPI

* Other molecular considerations may include intrinsic subtype by CNV, B-allele frequency, and RB-LOH.

TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

EVOLVE-BDT (real, enrolling)

ER+/HER2– and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurIST[17], same Bayesian arm-swap logic on daraxonrasib-combination arms.

> **Same architecture, different disease.**

What this demands of the statistical design

Four design axes any such trial needs.

Adaptive stopping	Real-time biomarker discovery	Longitudinal monitoring	In-trial evaluation
Bayesian futility / efficacy boundaries at each interim	Pre-specified signatures + agnostic discovery window	Serial ctDNA, imaging, biopsies on every patient	Biomarker-guided cohorts within the same protocol

Methods are ready. We’re operationalizing them in PDAC trials — starting with PANGEA.

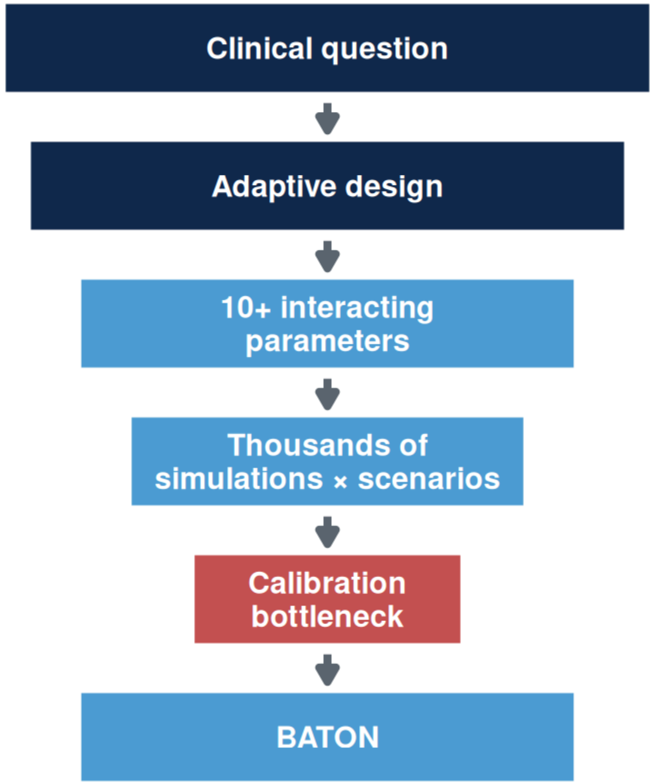
Putting these into practice

Why don’t we see more of these trials?

Why don't we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.



Methods adopted. Dose-optimization plans — where calibration cost bites — not[24]*.*

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

The calibration challenge: four competing goals

Designing an adaptive trial requires achieving all four simultaneously.

1. FDA compliance

Power ≥ 0.80
Type I ≤ 0.10

2. Minimize patients

Smaller E[N]
Efficient use of scarce patients

3. Stop bad early

Futility stopping
Protect patients from inactive arms

4. Accelerate good

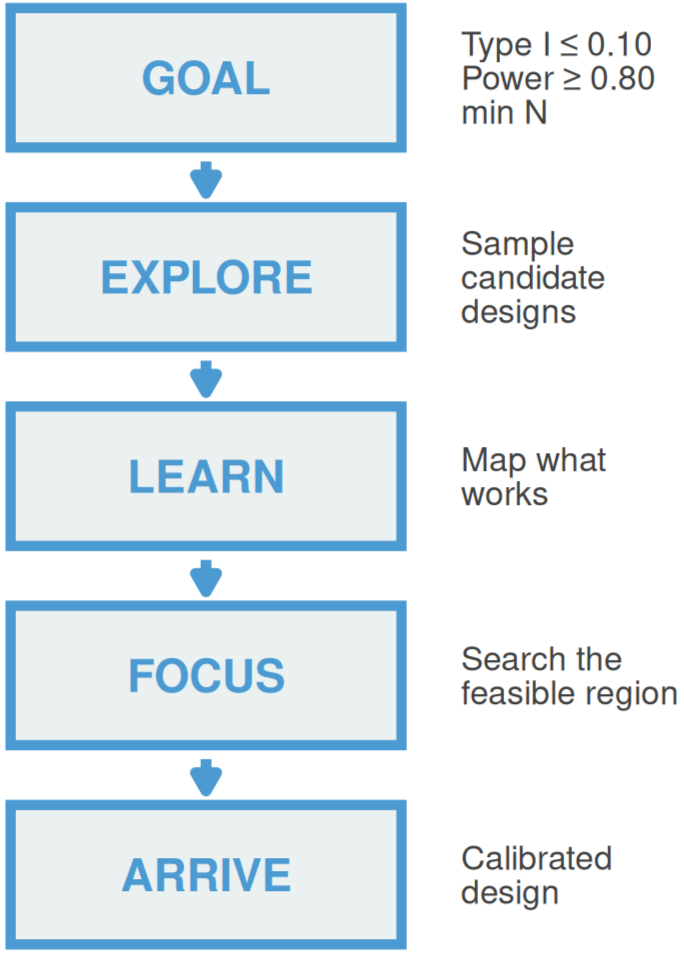
Efficacy stopping
Faster access when drug works

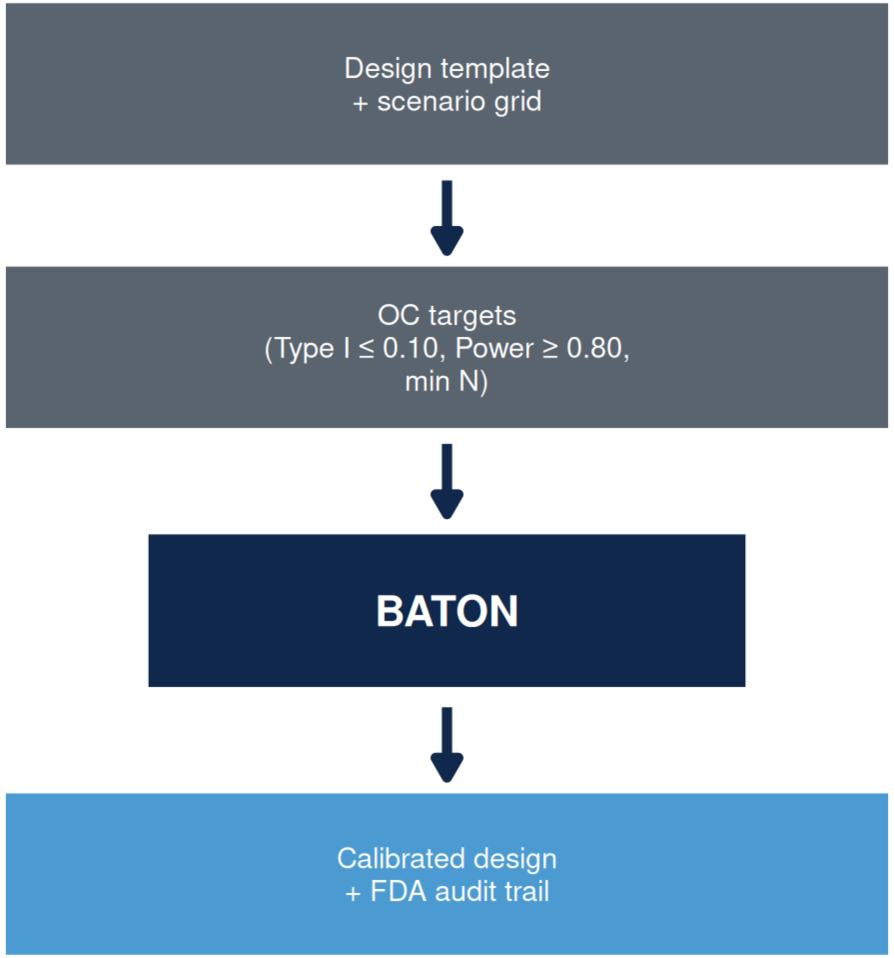
"We could design the science. We could not design the trial."

BATON: the calibration tool

BATON = **B**ayesian **A**daptive **T**rial **O**ptimizatio**N**[25]

BATON turns clinically necessary trial complexity into calibrated, launchable designs.





You bring the clinical inputs; BATON returns the calibration.

You bring the clinical design; BATON evaluates hundreds of candidates in silico and returns the ones that meet your targets. R package: github.com/naimurashid/BATON[25]

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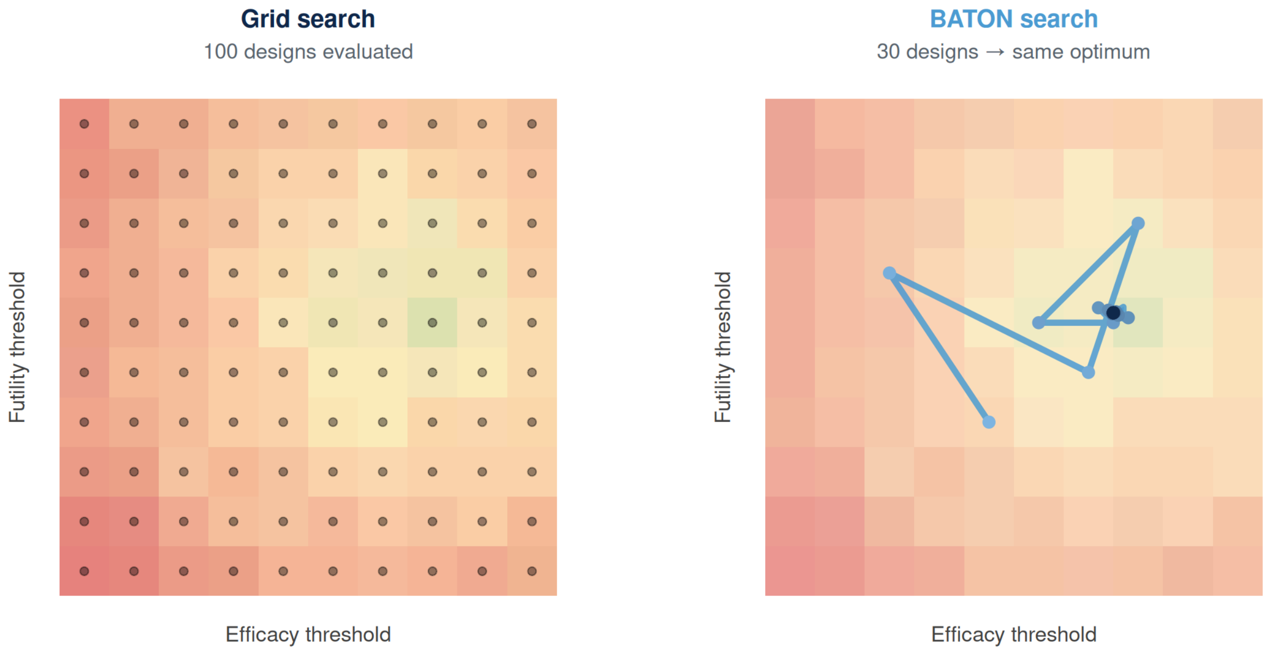
Grid search vs BATON

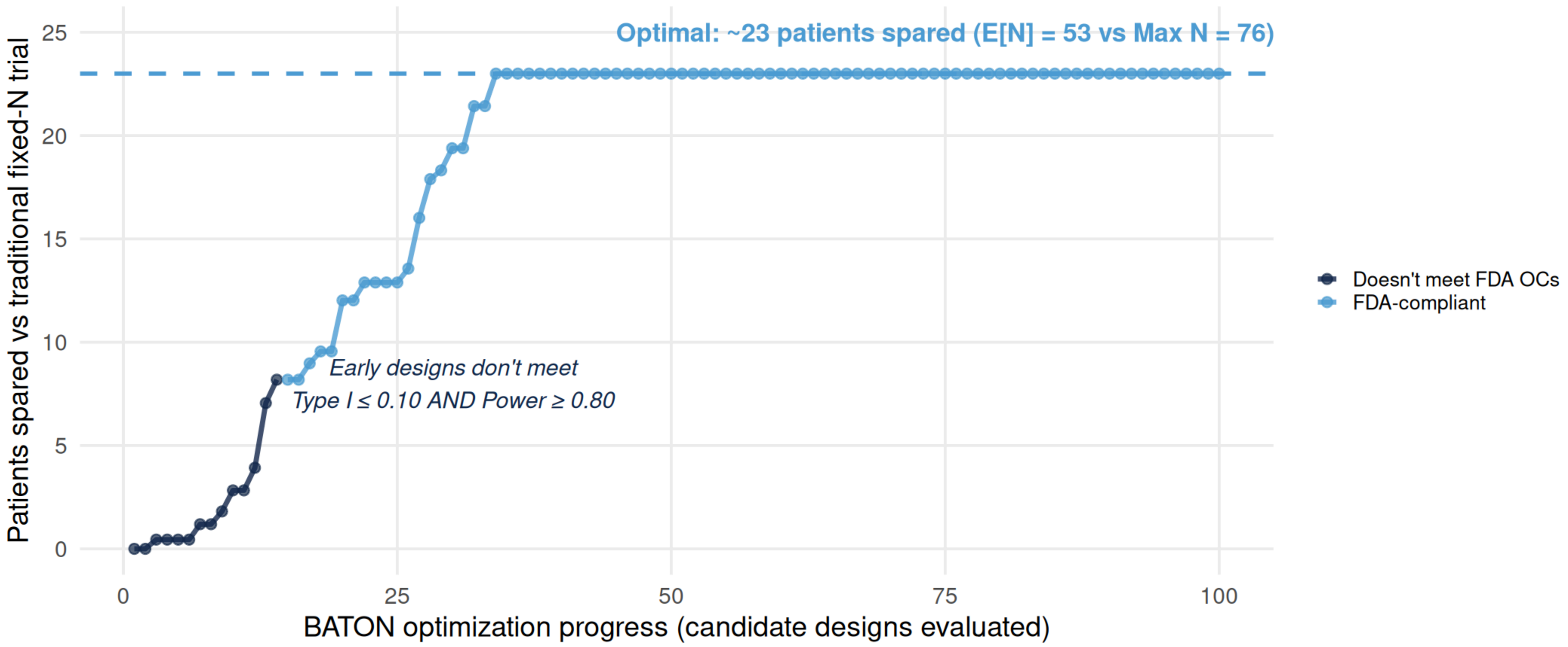


70% fewer evaluations — and the same optimum.

	MANUAL	BATON
Process	Trial-and-error	Systematic search
Wall-clock	Weeks to months	Hours
Output	One design, often under-validated	Full feasible region + FDA-grade audit trail

Same clinical question. Same statistical constraints. The bottleneck was the calibration.

BATON's output, framed as patients spared



~23

patients spared
(E[N] = 53 vs Max N = 76)

7.0%

achieved Type I
(target ≤ 10%)

82.2%

achieved Power
(target ≥ 80%)

Savings come from adaptive stopping; BATON's job is finding the design that achieves them within the OC constraints.

Each point on the convergence plot = one candidate design evaluated (≈10,000 simulated trials per design).

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurIST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

Sooner answer, fewer patients on a losing arm — without losing power.

PANGEA Phase II: calibrated by BATON

- After Phase I OBD → **Phase II Bayesian adaptive RCT:** GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)[26]
- **BATON-calibrated** four parameters under Type I ≤ 0.10, Power ≥ 0.80[25]

Achieved operating characteristics:

METRIC	VALUE
Power	**82.2%**
Type I error	7.0%
Max N	**76** (82 accrual w/ buffer)
Expected N under H_0	53.0
Expected N under H_1	58.1
Pr(early futility stop | H_0)	**77.9%**
Pr(early efficacy stop | H_1)	**81.4%**
Expected duration	27.3 mo

H_0 = drug inactive (no PFS benefit over GnP alone) · H_1 = drug active at the assumed effect size (HR 0.58)

> **Without BATON:** four interacting parameters rarely hit both constraints simultaneously. **With BATON:** joint calibration against both constraints, ~200 evaluations.

An unexpected finding: *caps set too low suppress early-efficacy declarations — even when the drug works, the trial waits for the final analysis to say so. BATON returned the* ***best trade-off curve*** *across candidate caps; we chose 76 to preserve early-efficacy stopping. Manual calibration would have anchored on the first feasible design.*

Where this is going

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:

PDAC RAS-backbone combinations OBD selection	**CRC** Joint-surface novel-novel doublets
Basket / umbrella Biomarker-stratified subtrials shared decision rules	**Robust calibration** Designs robust to uncertain trial assumptions

BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.

The new GI oncology trial template

DOSE

OBD selection,
not MTD
(U-BOIN, BOIN12)

→

COMBINE

Joint-surface
designs for RAS
combinations

→

ADAPT

Resistance-aware
longitudinal
monitoring

→

CALIBRATE

Hours, not
months — with
FDA audit trail

Each piece is mature methodology. The gap was infrastructure — and we're closing it.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs handle all three.
3. **BATON makes them practical to calibrate and launch.**

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Questions